Shengyuan Yang, Ph.D. **US Citizen**
Associate Professor, Department of Mechanical and Civil Engineering, Florida Institute of Technology (Florida Tech), Melbourne, Florida
Director, Micro and Nano Electro Mechanical Systems (MEMS/NEMS) and Cell Mechanics Laboratory
National Science Foundation (NSF) CAREER Awardee
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EDUCATION

Department of Mechanical Science and Engineering, **University of Illinois at Urbana-Champaign**, **Ph.D. in Mechanical Engineering**, Jul. 2007
Dissertation Title: "Micromachined Sensors and Their Applications in the Study of Mechanical Response of Single Living Cells"

Department of Precision Machinery and Instrumentation, **University of Science and Technology of China**, Hefei, China, **Bachelor of Engineering**, Jul. 1993
Major: Precision Machinery and Instrumentation; Minor: Electronic Technology and Computer Applications

PROFESSIONAL EXPERIENCE

Associate Professor, Department of Mechanical and Civil Engineering, *Florida Institute of Technology*, Melbourne, Florida, Mar. 2018 – present
Director, MEMS/NEMS and Cell Mechanics Laboratory

Visiting Scholar, Nanomechanics Laboratory, MIT (*Massachusetts Institute of Technology*), Cambridge, Massachusetts, Sep. – Dec. 2019

Visiting Associate Professor, Department of Engineering Mechanics, Zhejiang University, Hangzhou, China, Jul. – Aug. 2019

Assistant Professor, Department of Mechanical and Aerospace Engineering, *Florida Institute of Technology*, Melbourne, Florida, Aug. 2007 – Feb. 2018
Director, MEMS/NEMS and Cell Mechanics Laboratory

Laboratory Instructor, Summer Course on Cell Mechano-sensitivity: Theory and Hands-on Experiments with Nanofabricated Devices, Jul. 30 - Aug. 3, 2007, *University of Illinois at Urbana-Champaign*

Laboratory Instructor, GEM^4 (Global Enterprise for Micro-Mechanics and Molecular Medicine) Summer School on Cell and Molecular Mechanics in Biomedicine with a focus on infectious diseases, Aug. 7 - 18, 2006, MIT, Cambridge, Massachusetts

- Designed and taught the laboratory session: "Bio-MEMS based force sensors for the study of mechanical behavior of human red blood cells".

Research Fellow, Precision Engineering and Nanotechnology Center, *Nanyang Technological*

University, **Singapore**, Dec. 1998 – Jul. 2002

Research Associate, Department of Mechanical System Engineering, *Tokyo University of Agriculture and Technology*, Tokyo, **Japan**, Sep. 1998 – Nov. 1998

Have Taught the Following Courses at Florida Tech

Undergraduate level:

MEE 2024, Solids Modeling and 3-D Mechanical Design Principles;
MEE 2081, Applied Mechanics: Statics; MEE 2082, Applied Mechanics: Dynamics
AEE 3083, Mechanics of Materials; MEE 3090, Design of Machine Elements
MEE 3091, Theory of Machines

Graduate level:

MEE 5410, Elasticity; MEE 5460, Fracture Mechanics and Fatigue of Materials
MAE/BME 5740, Cellular Biomechanics

ADVISORS (for doctoral students)

Sang Joo Lee, Ph.D. student (graduated in Jun. 2016)
Dissertation Title: "Micro glass ball embedded gels to study cell mechanobiological responses to substrate curvatures and local substrate stiffness"

Mohamed Amro Waregh, Ph.D. student (graduated in 2021)
Dissertation Title: "On-line, real-time condition monitoring and fault diagnostics of gear-train systems"

SERVICES

Proposal Review Panelist, Division of Civil Mechanical and Manufacturing Innovation, Directorate for Engineering, **NSF** (2014 – present)

Proposal Review Panelist, SBIR/STTR, Division of Industrial Innovation and Partnerships, **NSF** (2016 – present)

Topic Editor, *Micromachines* (2072-666X)

Panel Fellow Cohort for the Game Changer Academies for Advancing Research Innovation, Directorate for Engineering, **NSF**, 2021

Professional Development Activities

Completed the **NSF Innovation Corps (I-Corps) Program**, Jan. – Mar. 2016, Georgia Institute of Technology

Attended the **Innovating Curriculum with Entrepreneurial Mindset Workshop** provided by Kern Entrepreneurial Engineering Network (KEEN), Melbourne, Florida, Mar. 7 - 9, 2016

Attending the Panel Fellow Cohort development program of the **Game Changer Academies for Advancing Research Innovation**, orginzied by CMMI, Directorate for Engineering, NSF, April-October, 2021.

PATENTS

1. <u>Shengyuan Yang</u>, "Methods and kits for cell sorting," United States Patent No.: US 10,871,435, Date of Patent: Dec. 22, 2020.

2. Shengyuan Yang, "Methods and kits for directing cell attachment and spreading," United States Patent No.: US 10,738,276, Date of Patent: Aug. 11, 2020.
3. Shengyuan Yang, "Micro and nano scale structures disposed in a material so as to present micrometer and nanometer scale curvature and stiffness patterns for use in cell and tissue culturing and in other surface and interface applications," United States Patent No.: US 9,512,397, Date of Patent: Dec. 6, 2016.
4. Shengyuan Yang, "Versatile, flexible, and robust MEMS/NEMS sensor for decoupled measuring of three-dimensional forces in air or liquids," United States Patent No.: US 9,021,897, Date of Patent: May 5, 2015.
5. Shengyuan Yang, "Micro and nano glass balls embedded in a gel presenting micrometer and nanometer scale curvature and stiffness patterns for use in cell and tissue culturing and a method for making same," United States Patent No.: US 8,802,430, Date of Patent: Aug. 12, 2014.
6. Shengyuan Yang, "Substrates with micrometer and nanometer scale stiffness patterns and curvature patterns," China Utility Mode Patent No.: ZL 2013 2 0124977.5, Date of Patent: Dec. 11, 2013.
7. Shengyuan Yang, "Methods and kits for guided stem cell differentiation," United States Patent Pending, Application No.: 15652938, Date of Application: Jul. 18, 2017.
8. Shengyuan Yang, "Curvature-defined concave and convex PDMS surfaces for use in cell and tissue culturing and in other surface and interface applications," United States Patent Pending, Application No.: 16713644, Date of Application: Dec. 13, 2019.
9. Shengyuan Yang, "Curvature-defined convex and concave gel surfaces for use in cell and tissue culturing and in other surface and interface applications," United States Patent Pending, Application No.: 16713756, Date of Application: Dec. 13, 2019.
10. Shengyuan Yang, "Methods of using curvature-defined surfaces with varying curvatures to direct cell attachment, spreading, and migration," United States Patent Pending, Application No.: 16713889, Date of Application: Dec. 13, 2019.

EXAMPLE PUBLICATIONS

1. **(Book)** Shengyuan Yang, *Micromachined Sensors to Study Cell Mechanical Response*, VDM Verlag Dr. Müller, Saarbrücken, Germany, 2008.
2. Shengyuan Yang, "Curvature-defined culturing technology – A new tool for the studies of stem cell mechanobiology: A perspective," *Stem Cell Research & Therapeutics*, Vol. 5, No. 1, 2020, pp. 146-163 (an invited review and perspective paper).
3. Sang Joo Lee and Shengyuan Yang, "Substrate curvature restricts spreading and induces differentiation of human mesenchymal stem cells," *Biotechnology Journal*, Vol. 12, No. 9, 2017, Art. No. 1700360.
4. Sang Joo Lee and Shengyuan Yang, "Micro glass ball embedded gels to study cell mechanobiological responses to substrate curvatures," *Review of Scientific Instruments*, Vol. 83, No. 9, 2012, Art. No. 094302.
5. Scott Siechen, Shengyuan Yang, Akira Chiba, and Taher Saif, "Mechanical tension contributes to clustering of neurotransmitter vesicles at presynaptic terminals," *Proceedings of the National Academy of Sciences of the United States of America (PNAS)*, Vol. 106, No. 31, 2009, pp. 12611-12616.

References will be available upon request.